Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	C. Morse	NDS 182, 167 (2022)	14-Sep-2021

Q(β^-)=−3217 *SY*; S(n)=5523 *SY*; S(p)=3793 *SY*; Q(α)=9.95×10^3 *7* 2021Wa16
ΔQ(β^-)=929, ΔS(n)=918, ΔS(p)=866 (2021WA16).
S(2n)=12606 SY *801*, S(2p)=6482 SY *866* (2021WA16).

^{289}Fl has been reported as the product of the ^{244}Pu(^{48}Ca,3n) reaction at JINR (2000OG05,2004OG07) and GSI (2010DU06,2011GA19); as the α-decay daughter of ^{293}Lv at JINR (2017KA66), at GSI (2012HO12), and at RIKEN (2017KA66); and in a chemistry experiment using a gas chromatography system at GSI (2014YA33). In all cases, ^{289}Fl was identified as a member of a chain of α-decaying nuclei terminated by spontaneous fission. Identification of the isotopes in the chains was based on the comparison of the observed decay properties with those of previously known nuclei.

Note: 2000OG05, 2001OG01, and 2002OG09 identify ^{289}Fl as ^{288}Fl, but this assignment was later revised.

Other: 1999OG10 reports one chain assigned to ^{289}Fl, but the decay properties of this chain have not been reproduced in subsequent publications.

Half-lives, branching ratios, and α-decay energies in this evaluation have been computed from the individual events listed in the references above. Half-life uncertainties have been computed according to the method of 1984SC13. An additional 10 keV systematic uncertainty is assumed for the α-decay energies, which is added in quadrature to the averaged statistical uncertainty.

^{289}Fl Levels

Cross Reference (XREF) Flags

A ^{293}Lv α decay (95 ms)

E(level)	$T_{1/2}$	XREF	Comments
0	2.4 s *+8−5*	A	%α=100; %SF<6 E(level): Assumed ground state. $T_{1/2}$: From 15 events.